



Think of placing a tiny paddlewheel at a point in the flow. The fluid pushes on each blade. If the flow is faster or differently directed on one side than the other, the imbalance creates a torque and the wheel spins.

- **Positive curl** ( $\nabla \times \mathbf{F} > 0$  in 2D): paddlewheel spins **CCW**. Right-hand rule: thumb out of page ( $+\hat{\mathbf{k}}$ ).
- **Negative curl** ( $\nabla \times \mathbf{F} < 0$  in 2D): paddlewheel spins **CW**. Right-hand rule: thumb into page ( $-\hat{\mathbf{k}}$ ).
- **Zero curl**: paddlewheel doesn't spin, even if fluid is moving fast.

**Key insight:** curl measures *local spin*, not global shape. A field can have curved streamlines but zero curl, and *straight* streamlines but nonzero curl!

**Example — shear flow:** In  $\mathbf{F} = \langle y, 0 \rangle$  every arrow is horizontal, yet the paddlewheel spins. The top blade ( $y > 0$ ) is pushed right; the bottom blade ( $y < 0$ ) is pushed left. This imbalance  $\implies$  **CW** rotation, and indeed  $Q_x - P_y = 0 - 1 = -1 < 0$ . Negative curl despite no circular streamlines.

In 3D,  $\nabla \times \mathbf{F}$  is a vector: its **direction** is the axis of fastest rotation (right-hand rule) and its **magnitude** is the rate of rotation.

### Example 1: Determining Curl Sign from the Paddlewheel

For each field, predict the paddlewheel behavior at the origin, then verify by computing the 2D curl.

- (a)  $\mathbf{F} = \langle 0, x \rangle$
- (b)  $\mathbf{F} = \langle -y, 0 \rangle$
- (c)  $\mathbf{F} = \langle y, x \rangle$

## 2D Reduction

For  $\mathbf{F} = \langle P, Q \rangle$  (2D, so  $R = 0$ ), only the  $\hat{\mathbf{k}}$  component survives:  $\nabla \times \mathbf{F} = (Q_x - P_y) \hat{\mathbf{k}}$ . We call the scalar  $Q_x - P_y$  the **2D curl** — exactly the integrand in Green's Theorem!

### Example 2: 3D Curl Computation

Find  $\nabla \times \mathbf{F}$  for  $\mathbf{F} = \langle xz, xyz, -y^2 \rangle$ .

### Example 3: 2D Curl: Rotation vs. Source

Compute the 2D curl of (a)  $\mathbf{F} = \langle -y, x \rangle$  and (b)  $\mathbf{F} = \langle x, y \rangle$ .

## 2. Divergence: Measuring Expansion

### Definition 2: Divergence of a Vector Field

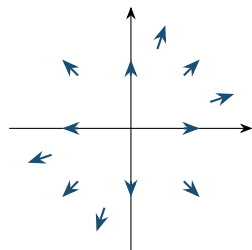
For  $\mathbf{F} = \langle P, Q, R \rangle$ , the **divergence** is (same thing, different notation):

$$\operatorname{div} \mathbf{F} = \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \underline{\hspace{2cm}}$$

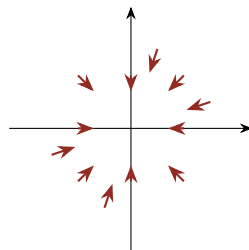
Just add up how much each component changes along its own direction.

- **Interpretation:** measures the net  $\underline{\hspace{2cm}}$
- $\operatorname{div} \mathbf{F} > 0$ : **source** (fluid spreading out)     $\operatorname{div} \mathbf{F} < 0$ : **sink** (fluid compressing)     $\operatorname{div} \mathbf{F} = 0$ : **incompressible**

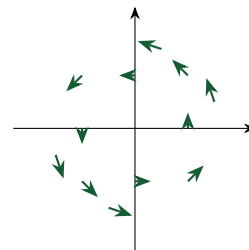
### Visualizing Divergence



**Source** ( $\operatorname{div} > 0$ )  
 $\langle x, y \rangle$ ,  $\operatorname{div} = 2$



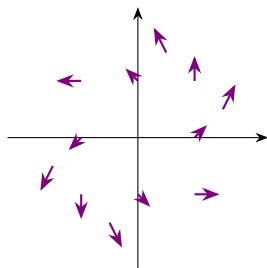
**Sink** ( $\operatorname{div} < 0$ )  
 $\langle -x, -y \rangle$ ,  $\operatorname{div} = -2$



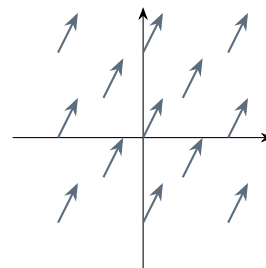
**Incompressible** ( $\operatorname{div} = 0$ )  
 $\langle -y, x \rangle$ ,  $\operatorname{div} = 0$

### Both Properties / Neither

Curl and divergence are **independent**. A field can have any combination:



**Both** ( $\operatorname{curl} \neq 0$ ,  $\operatorname{div} \neq 0$ )  
 $\langle x - y, x + y \rangle$   
 $\operatorname{curl} = 2$ ,  $\operatorname{div} = 2$



**Neither** ( $\operatorname{curl} = 0$ ,  $\operatorname{div} = 0$ )  
 $\langle 1, 2 \rangle$   
 Uniform flow

### Example 4: Incompressible Pipe Flow

A velocity field in a pipe is  $\mathbf{F} = \langle 2x, -y, -z \rangle$ . Is the flow incompressible?

### Example 5: Classifying a Field

For  $\mathbf{F} = \langle x, y, z \rangle$ , find  $\operatorname{div} \mathbf{F}$  and  $\nabla \times \mathbf{F}$ . Interpret.

### Example 6: Another Classification

For  $\mathbf{F} = \langle -y, x, 0 \rangle$ , find  $\operatorname{div} \mathbf{F}$  and  $\nabla \times \mathbf{F}$ . Interpret.

## 3. Key Identities

### Vector Calculus Identities

1.  $\nabla \times (\nabla f) = \underline{\hspace{2cm}}$  — conservative fields are always irrotational (by Clairaut's:  $f_{xy} = f_{yx}$ )  
*Contrapositive:* if  $\nabla \times \mathbf{F} \neq \mathbf{0}$ , then  $\mathbf{F}$  is **not** conservative.
2.  $\operatorname{div}(\nabla \times \mathbf{F}) = \underline{\hspace{2cm}}$  — curl fields are always divergence-free (Clairaut's again)
3. **Laplacian:**  $\nabla^2 f = \operatorname{div}(\nabla f) = \underline{\hspace{2cm}}$  — divergence of the gradient
4. **Product rules:**  
 $\nabla(fg) = f\nabla g + g\nabla f$        $\operatorname{div}(f\mathbf{F}) = f \operatorname{div} \mathbf{F} + \mathbf{F} \cdot \nabla f$        $\nabla \times (f\mathbf{F}) = f \nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$

### Operator Relationships

$$\text{Scalar field } f \xrightarrow{\text{grad}} \text{Vector field } \mathbf{F} \xrightarrow{\text{curl}} \text{Vector field } \mathbf{G} \xrightarrow{\text{div}} \text{Scalar field } h$$

The two zero identities say:  $\text{grad} \rightarrow \text{curl} = \mathbf{0}$ , and  $\text{curl} \rightarrow \text{div} = 0$ . In other words, the “derivative of a derivative” is zero in this chain.

### Example 7: Verifying $\nabla \times (\nabla f) = \mathbf{0}$

Let  $f(x, y, z) = x^2yz + \sin(yz)$ . Verify  $\nabla \times (\nabla f) = \mathbf{0}$ .

### Example 8: Proving $\operatorname{div}(\nabla \times \mathbf{F}) = 0$

Show that  $\operatorname{div}(\nabla \times \mathbf{F}) = 0$  for any smooth  $\mathbf{F} = \langle P, Q, R \rangle$ .

## 4. Green's Theorem Reformulated

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### Green's Theorem via Curl

$$\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \iint_D (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} dA$$

The circulation of  $\mathbf{F}$  around  $C$  equals the total curl over  $D$ . This is the 2D special case of **Stokes' Theorem** (coming soon!).

### Practice Problems

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**Problem 1.** Compute  $\operatorname{div} \mathbf{F}$  and  $\nabla \times \mathbf{F}$  for  $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ .

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**Problem 2.** Show that  $\mathbf{F} = \langle yz, xz, xy \rangle$  is both irrotational and incompressible.

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**Problem 3.** Find  $\nabla \times \mathbf{F}$  and  $\operatorname{div} \mathbf{F}$  for  $\mathbf{F} = \langle y^2z, x^2z, x^2y \rangle$ .

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**Problem 4.** Compute  $\nabla \times \mathbf{F}$  for  $\mathbf{F} = \langle e^x \cos y, -e^x \sin y, 0 \rangle$ . What does the result tell you?

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**Problem 5.** Verify  $\operatorname{div}(f\mathbf{F}) = f \operatorname{div} \mathbf{F} + \mathbf{F} \cdot \nabla f$  for  $f = xy$  and  $\mathbf{F} = \langle 1, 1, 0 \rangle$ .

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**Problem 6.** Prove: if  $\mathbf{F} = \nabla f$  and  $f$  is harmonic ( $\nabla^2 f = 0$ ), then  $\mathbf{F}$  is both irrotational and incompressible.

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**Problem 7.** Without computing, predict whether the 2D curl is positive, negative, or zero at the origin. Then verify.

(a)  $\mathbf{F} = \langle -2y, 3x \rangle$       (b)  $\mathbf{F} = \langle y, y \rangle$       (c)  $\mathbf{F} = \langle x^2, -y^2 \rangle$

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**Problem 8.** The velocity field of a whirlpool is modeled by  $\mathbf{F} = \langle -y, x, 2z \rangle$ . Find  $\nabla \times \mathbf{F}$  and  $\operatorname{div} \mathbf{F}$ . Does this flow rotate, expand, or both?